

THE RESIDUAL ATLAS

A field guide to the inhabitants of the model universe

Compact Einstein-Maxwell universe					
Ordinary and additional gravitational waves on a closed spatial universe supported by magnetic flux.					
Inhabitant	Causal travel	Classical pairing	Nonlinear survival	Observer response	Quantum status
Additional fourth-order gravitational waves	Next frontier	Demonstrated	Limit found	Next frontier	Next frontier
Balanced ordinary + fourth-order wave	Next frontier	Demonstrated	Demonstrated	Next frontier	Next frontier
Einstein / ϕ / ω / extra / exceptional ℓ 11 nonzero k	Next frontier	Demonstrated	Next frontier	Bridge not built	Bridge not built
Einstein / ϕ / ω / interaction / abd times ℓ 12 extra	Next frontier	Demonstrated	Next frontier	Next frontier	Next frontier
Einstein / ϕ / ω / interaction / d times ℓ 12 extra	Next frontier	Demonstrated	Next frontier	Next frontier	Next frontier
Einstein / ϕ / ω / interaction / homogeneous twist times ℓ 12 extra	Next frontier	Demonstrated	Next frontier	Next frontier	Next frontier
Einstein / ϕ / ω / mixed / aligned twist ℓ 12 extra compatibility face	Next frontier	Demonstrated	Limit found	Next frontier	Next frontier
Exceptional fourth-order dipole	Next frontier	Demonstrated	Limit found	Next frontier	Next frontier
Homogeneous motion balanced by twist	Next frontier	Demonstrated	Demonstrated	Next frontier	Next frontier
Linear dictionary between Einstein and fourth-order gravity	Next frontier	Demonstrated	Next frontier	Bridge not built	Bridge not built
Ordinary Einstein-Maxwell gravitational waves	Next frontier	Demonstrated	Next frontier	Next frontier	Next frontier
Twist-balanced exceptional dipole	Next frontier	Demonstrated	Limit found	Next frontier	Next frontier

Spherical vacuum universe					
Free reduced cylinder modes and deformation classes; these are not yet asymptotic particles.					
Inhabitant	Causal travel	Classical pairing	Nonlinear survival	Observer response	Quantum status
Cylinder mode tower A, negative chirality	Demonstrated	Demonstrated	Next frontier	Bridge not built	Demonstrated
Cylinder mode tower A, positive chirality	Demonstrated	Demonstrated	Next frontier	Bridge not built	Demonstrated
Cylinder mode tower E, negative chirality	Demonstrated	Demonstrated	Next frontier	Bridge not built	Demonstrated
Cylinder mode tower E, positive chirality	Demonstrated	Demonstrated	Next frontier	Bridge not built	Demonstrated
Cylinder mode tower L, negative chirality	Demonstrated	Demonstrated	Next frontier	Bridge not built	Demonstrated
Cylinder mode tower L, positive chirality	Demonstrated	Demonstrated	Next frontier	Bridge not built	Demonstrated
Negative-chirality deformation class (not a particle)	Demonstrated	Demonstrated	Next frontier	Not part of this test	Limit found
Positive-chirality deformation class (not a particle)	Demonstrated	Demonstrated	Next frontier	Not part of this test	Limit found

Bridges between universes					
Only explicit carrier maps may identify inhabitants defined on different backgrounds.					
Inhabitant	Causal travel	Classical pairing	Nonlinear survival	Observer response	Quantum status
Bh / crosswalk / compact-cylinder	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Can a compact-universe obstruction reach the Berger detector?	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Classical / berger / crosswalk / retained36 to einstein extra	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Classical / crosswalk / bach flat parent to metric	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Classical tangent cone to interacting BRST	Bridge not built	Bridge not built	Demonstrated	Bridge not built	Bridge not built
Compact gravitational waves carried to other universes	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Curvature observable to particle guard	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Curved-background quantum ghost calculation to particle guard	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Euclidean determinant to particle guard	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Local anomaly class to particle guard	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Nonlinear / berger / crosswalk / retained36 to residual branches	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Observer / crosswalk / berger physical branch to detector	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Quantum / crosswalk / algebraic cubic weyl carrier to particle	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Quantum / crosswalk / cpt universal third curvature kernel to particle	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Quantum / crosswalk / curvature squared covariant log form factor to particle	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Quantum / crosswalk / flat tt log form factor to particle	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Quantum / crosswalk / fv conformedized c2 log form factor to particle	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Quantum / crosswalk / third curvature weyl carrier manifest to particle	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built

Berger clock universe					
Light, clocks, emitters and detectors on the rotating squashed-sphere laboratory.					
Inhabitant	Causal travel	Classical pairing	Nonlinear survival	Observer response	Quantum status
Berger quantum carrier bridge	Demonstrated	Demonstrated	Next frontier	Bridge not built	Limit found
Clock-oscillation bound for unresolved detector waves	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Detector response after nonlinear restrictions	Next frontier	Next frontier	Next frontier	Next frontier	Next frontier
Finite clock-driven detector propagation	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
First rigorous bound on unresolved detector waves	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Localized light-emitter preparations	Demonstrated	Demonstrated	Next frontier	Demonstrated	Next frontier
Observer / berger / detector profile / adaptive cutoff preflight	Next frontier	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / adaptive diagonal fraction scale	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / adaptive scalar stability	Limit found	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / correlated central clock power	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / correlated central scalar	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / correlated diagonal fraction stream	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / correlated extreme axial	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / correlated intermediate jacobi	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / correlated sobolev n1	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / green weighted spatial tail reduction	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / haar normalization repair	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / jacobi axial stability	Limit found	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / moving profile clock derivative tail	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / polarization scalar closure	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / selected charge block companion closure	Limit found	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / selected charge block form companion clock rail	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / selected charge block scalar companion completion	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / selected charge block temporal bandwidth preflight	Limit found	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / selected clock power polarized form	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile / selected p0 polarized form	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile tail / two j138 exact t input	Limit found	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / detector profile tail / two j4 cutoff	Limit found	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / temporal green / blockwise functional calculus	Demonstrated	Not part of this test	Not part of this test	Next frontier	Not part of this test
Observer / berger / temporal green / order14 two j138	Limit found	Not part of this test	Not part of this test	Next frontier	Not part of this test
Relational light and redshift signal	Demonstrated	Demonstrated	Next frontier	Demonstrated	Next frontier
Two independent detector responses	Demonstrated	Next frontier	Not part of this test	Next frontier	Next frontier

Black-hole exterior					
Static black holes, gauge directions and the still-open dynamical perturbation sectors.					
Inhabitant	Causal travel	Classical pairing	Nonlinear survival	Observer response	Quantum status
Additional axial black-hole waves	Next frontier	Next frontier	Next frontier	Next frontier	Next frontier
Bh / background / static-family	Next frontier	Demonstrated	Next frontier	Next frontier	Next frontier
Bh / bridge / compact-branch-comparison	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Ordinary axial black-hole waves	Next frontier	Demonstrated	Next frontier	Next frontier	Next frontier
Polar black-hole waves	Next frontier	Next frontier	Next frontier	Next frontier	Next frontier
Static black-hole parameter changes	Next frontier	Demonstrated	Next frontier	Next frontier	Next frontier
Time-dependent conformal gauge motion	Not part of this test	Demonstrated	Next frontier	Not part of this test	Next frontier
Time-dependent coordinate gauge motion	Not part of this test	Demonstrated	Next frontier	Not part of this test	Next frontier

New atlas entries					
New certified inhabitants not yet assigned a public presentation group.					
Inhabitant	Causal travel	Classical pairing	Nonlinear survival	Observer response	Quantum status
Causal gravity on the conformal Nariai family	Demonstrated	Demonstrated	Next frontier	Not part of this test	Next frontier
Causal parent complex on Bach-flat backgrounds	Demonstrated	Demonstrated	Next frontier	Not part of this test	Next frontier
Classical / berger / retained gravity clock maxwell	Demonstrated	Demonstrated	Demonstrated	Demonstrated	Next frontier
Classical / vacuum cylinder / deformation / w minus squared	Demonstrated	Demonstrated	Next frontier	Not part of this test	Next frontier
Classical / vacuum cylinder / deformation / w plus squared	Demonstrated	Demonstrated	Next frontier	Not part of this test	Next frontier
Classical / vacuum cylinder / one particle / a	Demonstrated	Demonstrated	Next frontier	Not part of this test	Next frontier
Classical / vacuum cylinder / one particle / e	Demonstrated	Demonstrated	Next frontier	Not part of this test	Next frontier
Classical / vacuum cylinder / one particle / l	Demonstrated	Demonstrated	Next frontier	Not part of this test	Next frontier
First causal variation away from the Nariai family	Demonstrated	Demonstrated	Next frontier	Not part of this test	Next frontier
Nonlinear / abstract / finite harmonic / tangent cone naturality	Not part of this test	Not part of this test	Demonstrated	Not part of this test	Not part of this test
Nonlinear / berger / bridge2 / invariant interaction to physical branches	Bridge not built	Bridge not built	Bridge not built	Bridge not built	Bridge not built
Nonlinear / berger / retained mixed ℓ 13 / filtered cyclic obstruction	Next frontier	Demonstrated	Limit found	Bridge not built	Next frontier
Nonlinear / product / axial polar einstein minus to axial extra / ℓ 12 m0 k0 sum frequency	Bridge not built	Next frontier	Demonstrated	Bridge not built	Bridge not built
Nonlinear / product / bridge1 / exceptional ℓ 11 k0 solution cofiber	Bridge not built	Demonstrated	Next frontier	Bridge not built	Bridge not built
Nonlinear / product / bridge1 / exceptional ℓ 11 nonzero k solution cofiber	Bridge not built	Demonstrated	Next frontier	Bridge not built	Bridge not built
Nonlinear / product / bridge1 / generic axial relative branch map	Bridge not built	Limit found	Next frontier	Bridge not built	Bridge not built
Nonlinear / product / bridge1 / generic polar relative branch map	Bridge not built	Limit found	Next frontier	Bridge not built	Bridge not built
Nonlinear / product / bridge1 / generic standard pairing cyclic map inertia obstruction	Not part of this test	Limit found	Bridge not built	Bridge not built	Bridge not built
Nonlinear / product / homogeneous twist times ℓ 12 extra / bounded tangent cone	Bridge not built	Demonstrated	Demonstrated	Bridge not built	Bridge not built
Nonlinear / product / homogeneous twist times ℓ 12 extra / complete bounded resonance matrix	Bridge not built	Demonstrated	Demonstrated	Bridge not built	Bridge not built
Nonlinear Einstein-to-fourth-order bridge through third order	Bridge not built	Limit found	Next frontier	Bridge not built	Bridge not built

Demonstrated Limit found Next frontier Not part of this test Bridge not built

Each verdict is restricted to its declared background, carrier, boundaries and charge sector.